

Processing Water Samples¹

water sample (150-300 mL; n =3-4)



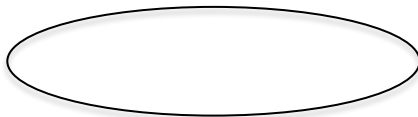
solids, liquid, microbes



sterilized Miracloth to capture solids



liquids, microbes



0.2 micron filter to capture microbes



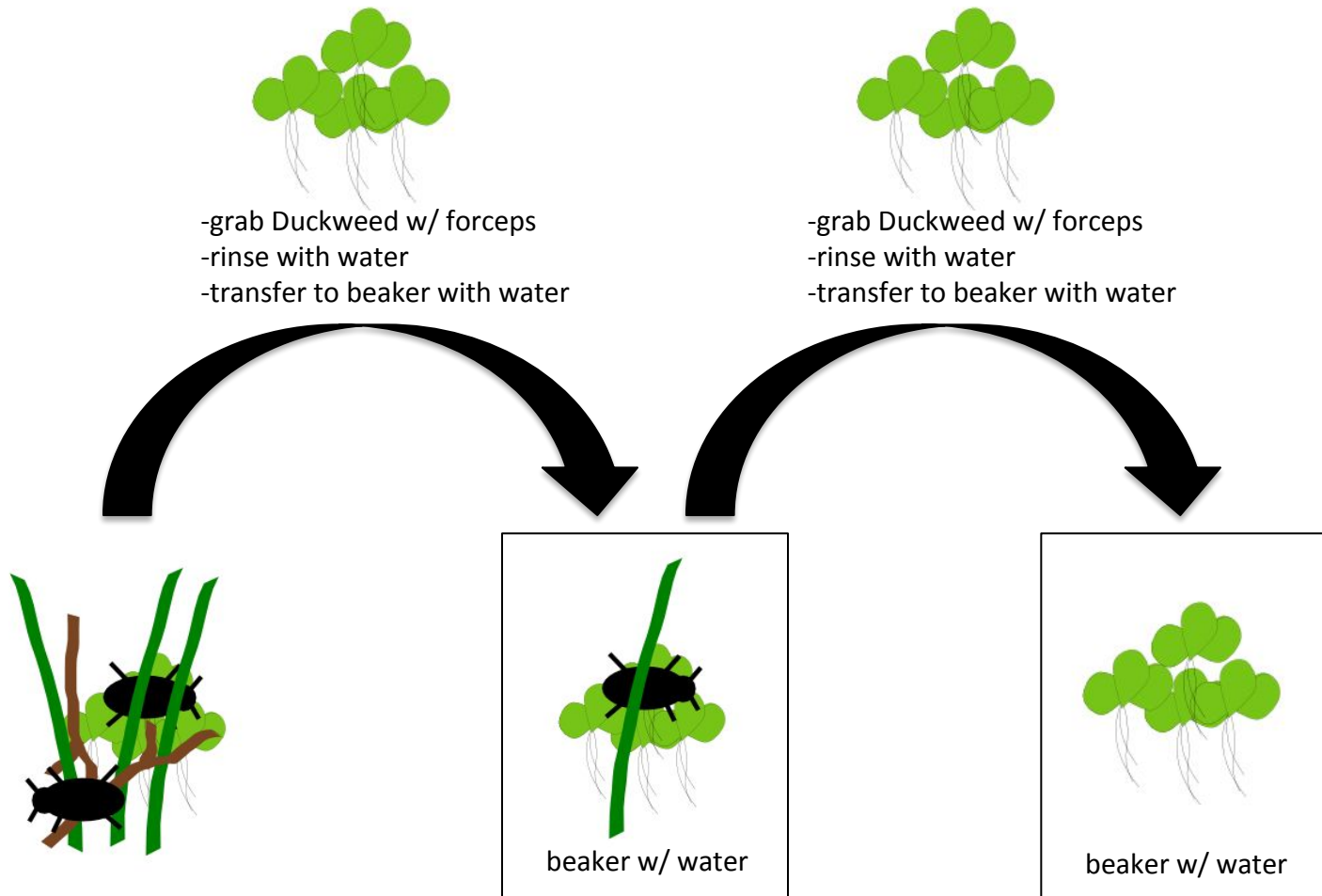
cut filter,
flash freeze,
and store
for 16S rRNA
amplicon
sequencing



liquid

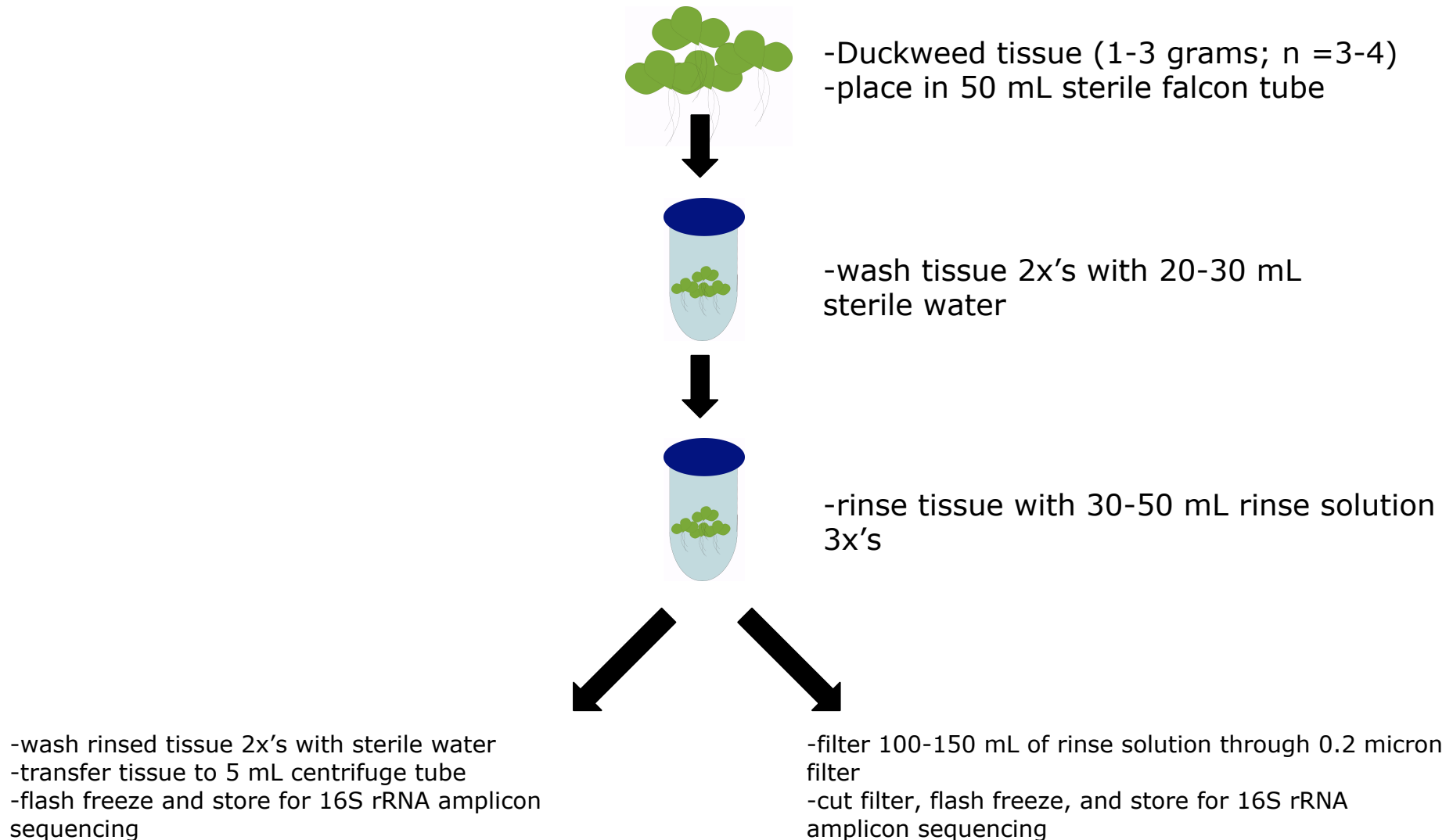
¹See "Processing Water Samples" protocol.

Separating Duckweed From Other Solids



Continue transfer to beakers until a pure Duckweed culture is obtained.

Processing Rinse Solution And Duckweed Tissue Samples²



²See "Processing Rinse Solution And Duckweed Tissue Samples" protocol.

Processing Water Samples

1. Get 150 mL 0.2 micron filter unit. Lay sterilized miracloth on top of filter unit.
2. Collect 150-300 mL of water sample and pour through sterilized Miracloth into filter unit.
3. Filter water sample through 0.2 micron filter unit by turning vaccum on.
4. Cut 0.2 micron filter from unit using sterile scalpel.
5. Place into 5 mL centrifuge tube.
6. Flash freeze and store at -80C until processing.

Processing Rinse Solution And Duckweed Tissue Samples

1. Collect 1-3 grams of fresh weight Duckweed from environment sample. (See "Separating Duckweed From Other Solids" to obtain pure Duckweed tissue from environment.)
2. Place into 50 mL falcon tube.
3. Wash tissue 2x's with sterile water.
4. Rinse tissue 3-5x's with 30-40 mL rinse solution. Filter rinse solution through 0.2 micron filter each time.
5. Filter a total of 100-150 mL of rinse solution through 0.2 micron filter.
6. Cut filter using sterile scalpel, transfer to 5 mL centrifuge tube, flash freeze, and store for 16S rRNA amplicon sequencing.
7. Wash rinsed Duckweed tissue 2x's with sterile water.
8. Transfer tissue to 5 mL centrifuge tube.
9. Flash freeze and store for 16S rRNA amplicon sequencing.